ARTIFICIAL INTELLIGENCE GROUP • UNIVERSITY OF HAGEN

Argumentation-based Causal Reasoning

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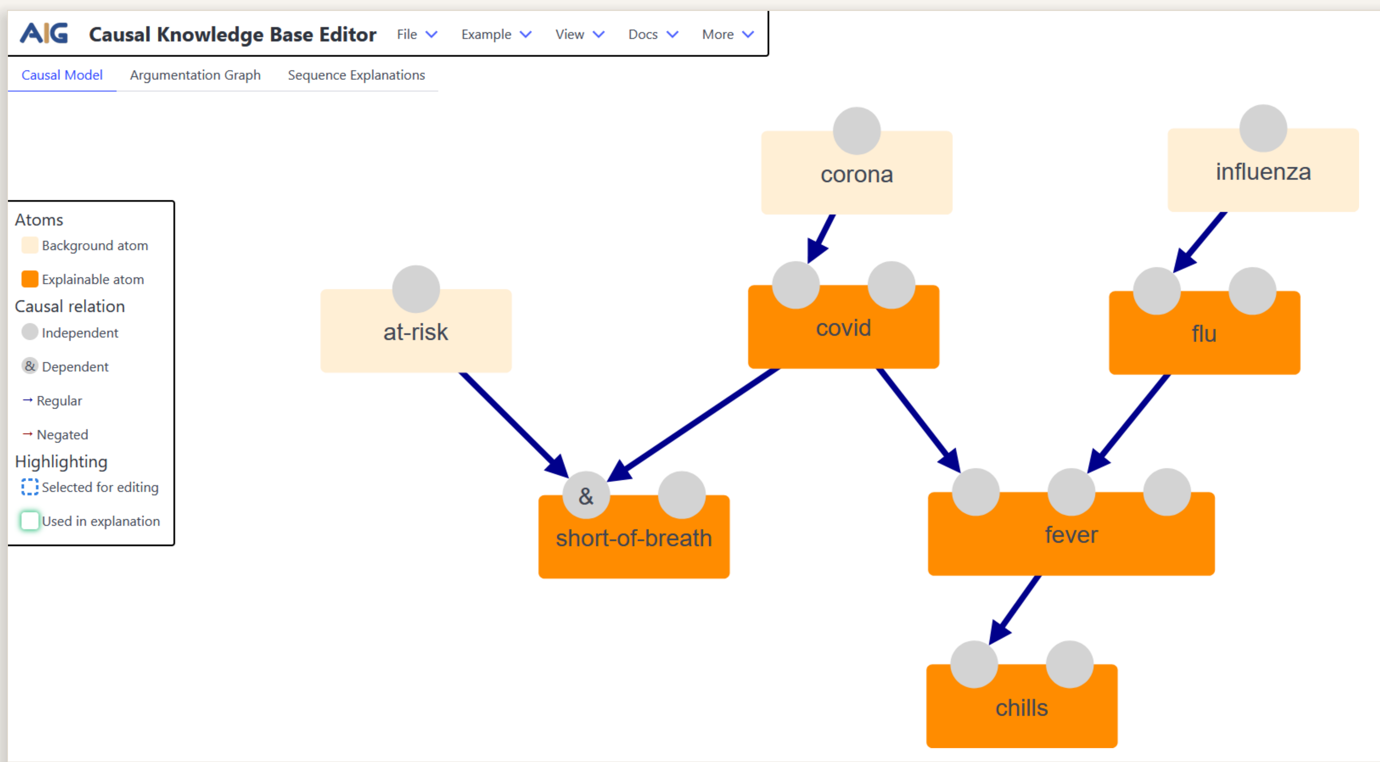
POWERED BY **TweetyProject** reasoning libraries & solvers

WEB TOOL • OPEN SOURCE

Build causal models and reason about **why**.

The CAUSAL KNOWLEDGE BASE EDITOR is a visual interface to construct causal knowledge bases, **reason** over them through their induced argumentation framework, and get **human-readable explanations**.

SCAN TO TRY IT LIVE
causal-knowledge-base-editor.aig.fernuni-hagen.de



Graphical editor — a causal model for diagnosing a viral disease, with structural equations and assumptions.

HOW IT WORKS

1 Causal model

Boolean-valued variables tied together by *structural equations* $v \leftrightarrow \phi$ — one causal mechanism per explainable atom — plus assumptions on the uncontrollable background atoms.

2 Induced AF

Arguments are minimal sets of assumptions that consistently entail a conclusion; an argument *undercuts* another when it negates one of its premises, yielding an argumentation framework.

3 Reason & explain

Given an observation, a conclusion holds exactly when it is in *every stable extension* of the induced AF — with dependency and sequence-based explanations.

FEATURES

Model

Construct causal models graphically — atoms, structural equations, automatic layout.

Assume & observe

Fix assumptions on background atoms and choose the observations to reason from.

Reason

Valid conclusions computed from the stable extensions of the induced AF.

Explain

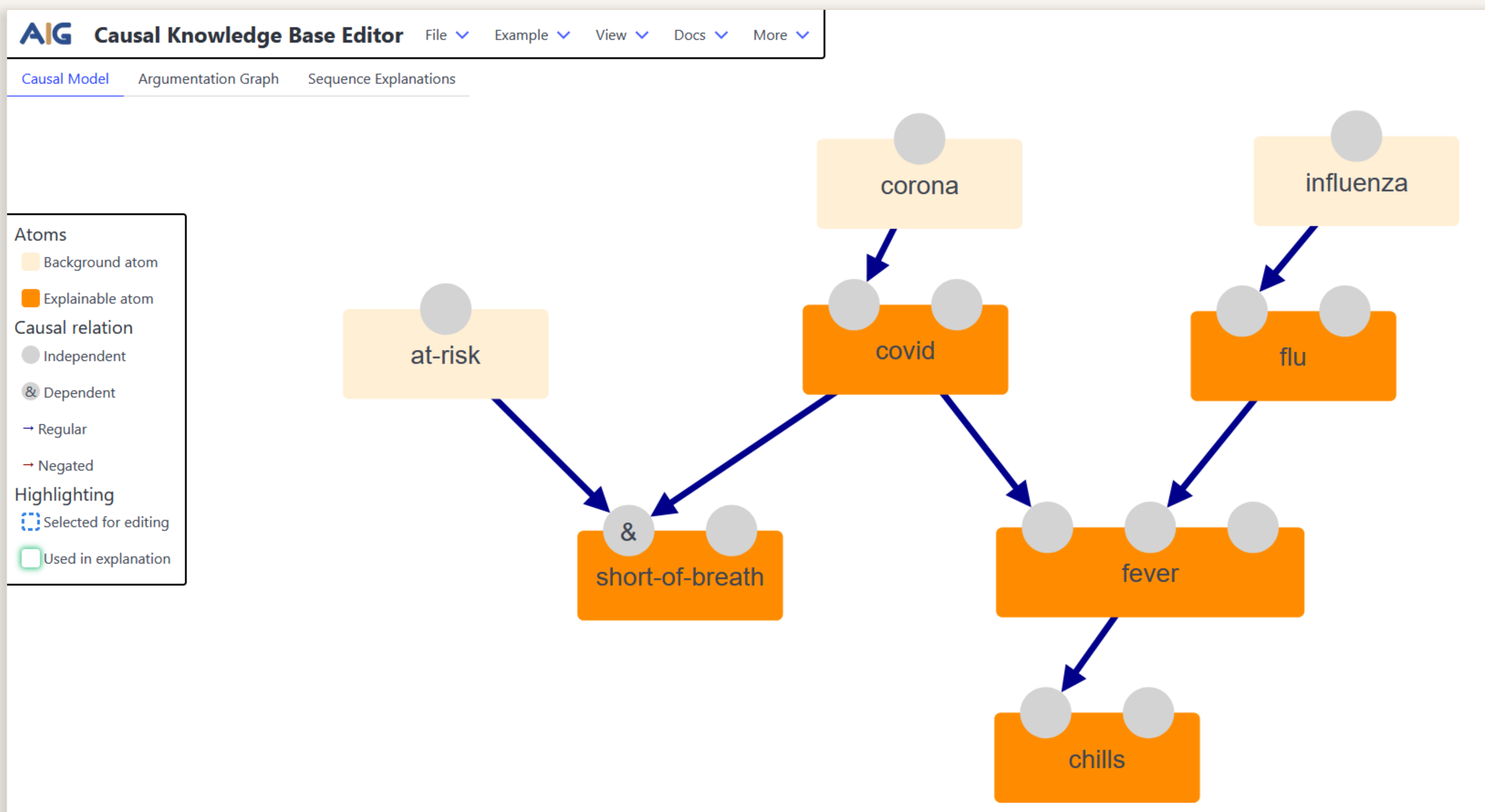
Dependency explanations and sequence-based explanations of conclusion arguments.

Switch views

Flip freely between the causal model and the induced AF behind it.

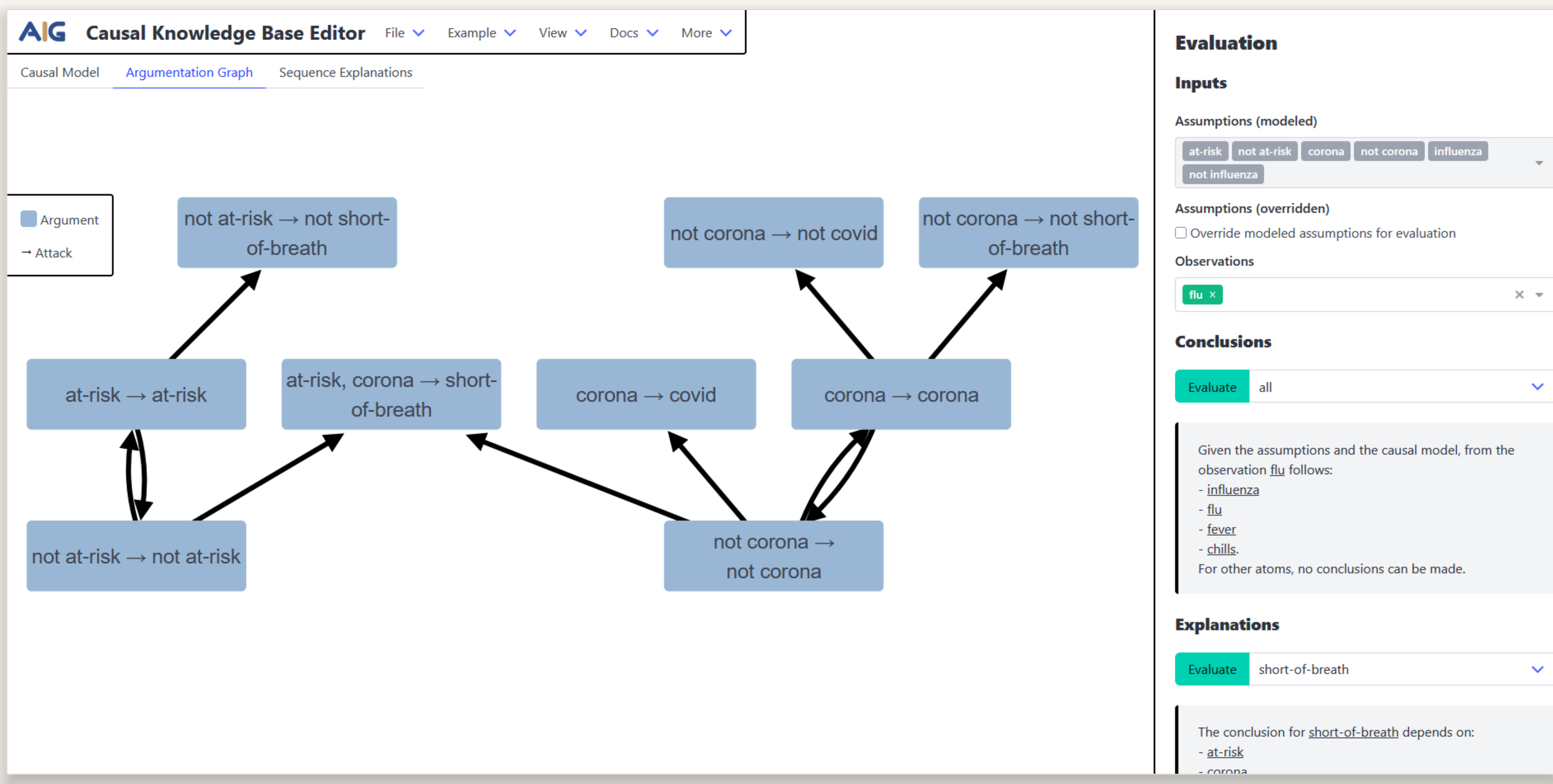
IN ACTION

Causal model



A causal model for diagnosing a viral disease — explainable atoms, their structural equations and the background assumptions.

Induced AF & evaluation



The argumentation framework induced from the model, with the valid conclusions read off its stable extensions.

Assumptions & observations

screenshot to come

The input panel where the user fixes background assumptions and observations before evaluating. *Screenshot to provide.*

Sequence-based explanation

screenshot to come

The step-by-step explanation of a conclusion argument. *Screenshot to provide.*

OUTLOOK

- richer presentation and explanation of results within the AF
- counterfactual reasoning via the twin-model method

UNDER THE HOOD

A JavaScript front end built on the open-source `aig-graph-component`; causal reasoning and AF semantics by *TweetyProject*.

Open source — graphical modelling made to be explored, extended and reused.